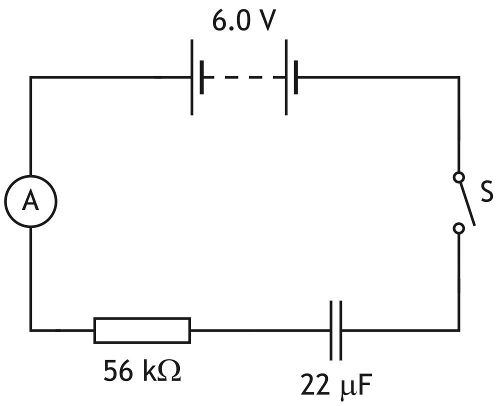
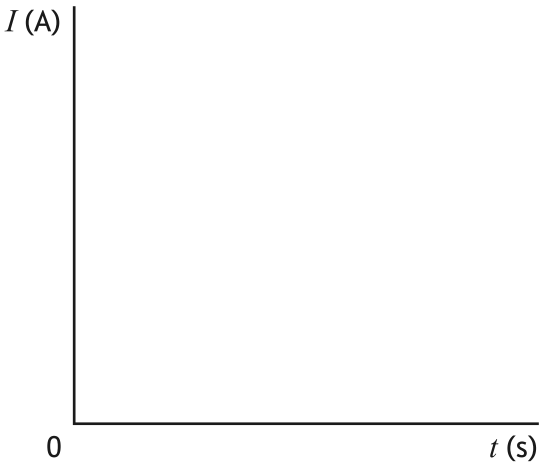
**12.** (a) A student sets up a circuit to investigate the charging of a capacitor.




The capacitor is initially uncharged. The student closes the switch and the capacitor takes 6.2 s to fully charge.

On the axes below, sketch a graph of current $I$ against time $t$ for the capacitor to fully charge.

Numerical values are required on both axes. **3**

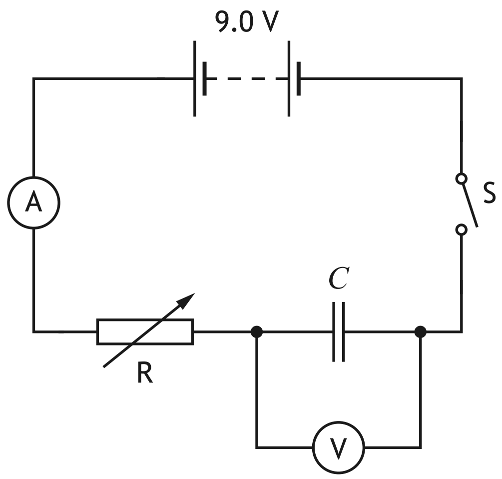



(An additional graph, if required, can be found on *page 52*.)

**[Turn over**

**12. (continued)**

(b) In a second experiment, the student sets up a circuit to determine the capacitance $C$ of a different capacitor.




The resistance of the variable resistor can be adjusted to maintain a constant charging current.

Initially, the capacitor is uncharged.

Switch S is closed and the capacitor begins to charge.

(i) State whether the student should be increasing or decreasing the resistance of the variable resistor R to maintain a constant charging current.

Justify your answer. **2**

(ii) During the experiment, there is a constant charging current of 15 $\mu$A.

The capacitor takes 28 s to fully charge.

Calculate the charge stored on the capacitor when it is fully charged. **3**

*Space for working and answer*

**12. (b) (continued)**

(iii) Calculate the capacitance $C$ of the capacitor. **3**

*Space for working and answer*

**[Turn over**